

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

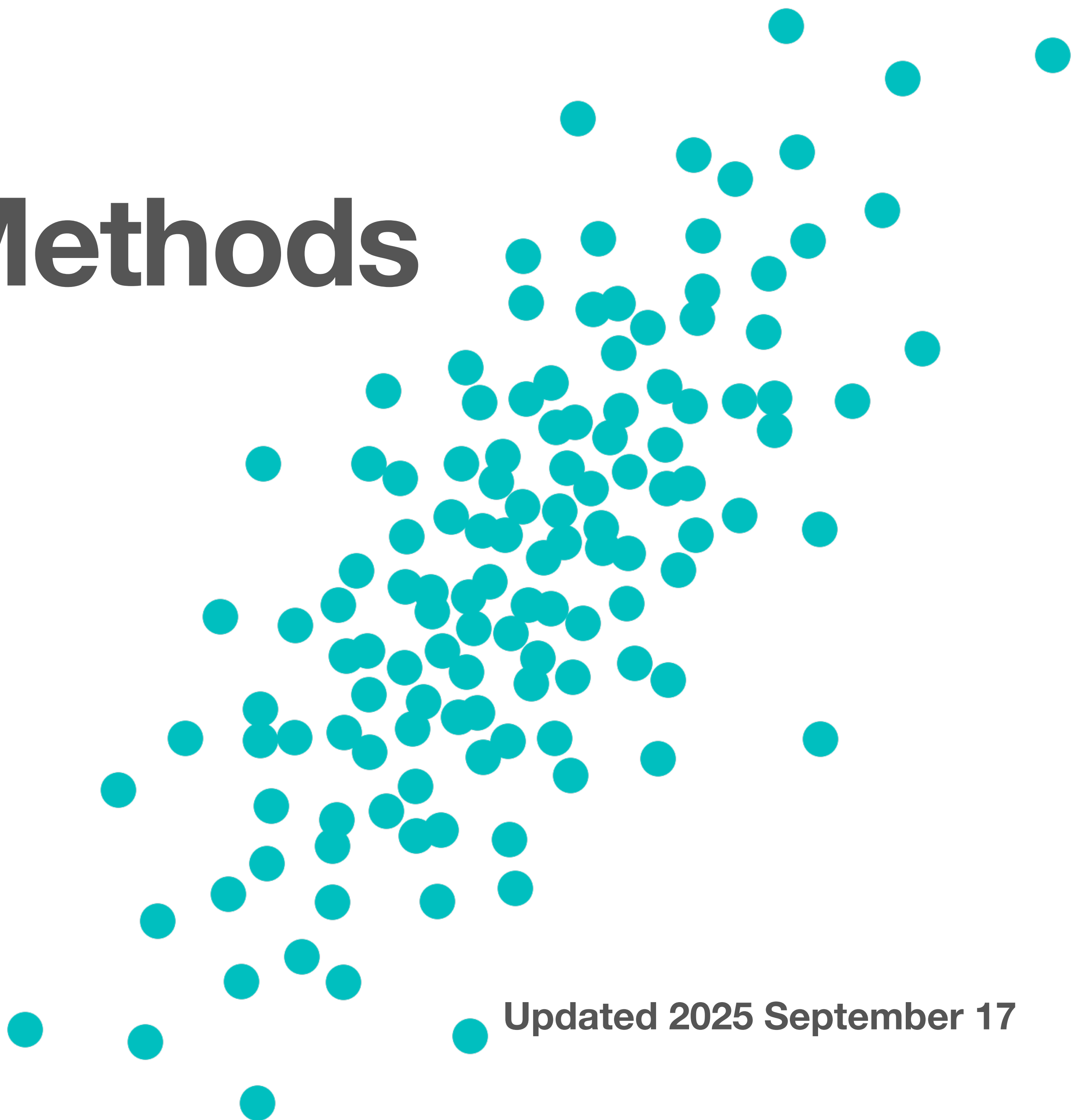
Improving Efficiency

Selected Topics

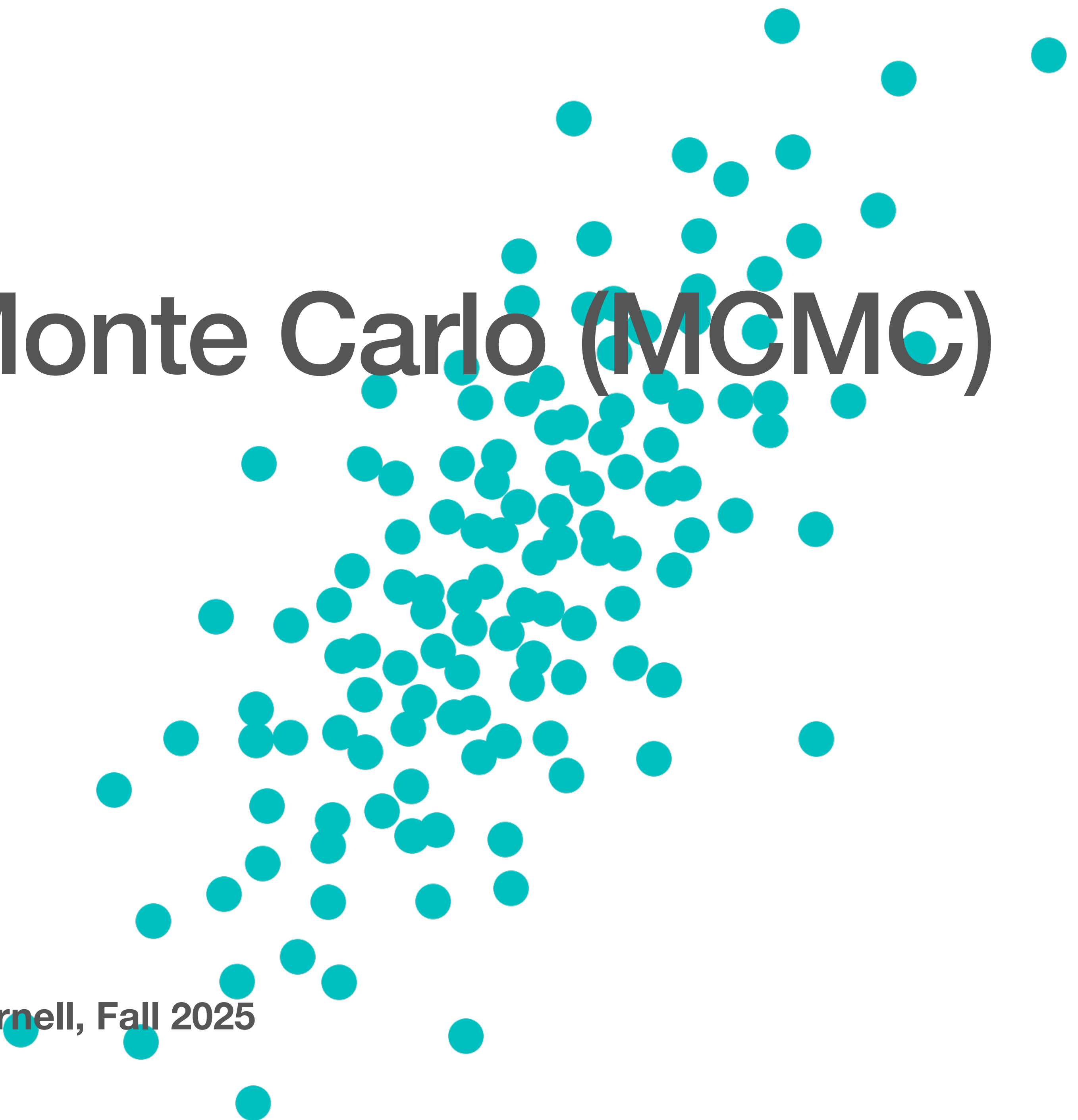
Git [website](#) and [repository](#)
[Canvas](#)

Fred Hickernell, Fall 2025

Updated 2025 September 17



Markov Chain Monte Carlo (MCMC)

A decorative graphic consisting of numerous teal-colored dots of varying sizes, scattered across the right side of the slide. The dots are more densely packed in the center-right area and become sparser towards the top and bottom right corners.

Owen, Chapters 11– 12

Assignment 3 due ??

Project Selection due Oct 1



How do we generate samples when the distribution is complicated?



How do we generate samples when the distribution is complicated?

- Inverse cumulative distribution function applied to uniform samples works for simple cases



How do we generate samples when the distribution is complicated?

- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians



How do we generate samples when the distribution is complicated?

- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling



How do we generate samples when the distribution is complicated?

- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling
 - Propose from a simple density that you know bounds



How do we generate samples when the distribution is complicated?

- Inverse cumulative distribution function applied to uniform samples works for simple cases
- Correlated multivariate Gaussian distributions are possible through affine transformations of univariate Gaussians
- Acceptance-rejection sampling
 - Propose from a simple density that you know bounds
 - An unnormalized target density



How do we generate samples when the distribution is complicated?

- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling
 - Propose from a simple density that you know bounds
 - An unnormalized target density
 - May reject a lot of samples



How do we generate samples when the distribution is complicated?

- **Inverse cumulative distribution** function applied to uniform samples works for simple cases
- **Correlated multivariate Gaussian** distributions are possible through affine transformations of univariate Gaussians
- **Acceptance-rejection** sampling
 - Propose from a simple density that you know bounds
 - An unnormalized target density
 - May reject a lot of samples
- **Markov chain Monte Carlo** (MCMC) moves samples toward areas of higher target density



Metropolis–Hastings algorithm

Suppose that

- You really **want to** sample X with known (unnormalized) **target** density q
- You have a **proposal** density $q_{\text{new|old}}$ to select the next point starting from where you are

Given $X[0]$

For $i = 0$ to $n-1$

Generate $Z \sim q_{\text{new|old}}(\cdot | X_i)$, $U \sim \mathcal{U}[0,1]$

If $U \leq \min\left(1, \frac{q(Z)q_{\text{new|old}}(X_i | Z)}{q(X_i)q_{\text{new|old}}(Z | X_i)}\right) = \min\left(1, \frac{\text{joint density of first } Z \text{ then } X_i}{\text{joint density of first } X_i \text{ then } Z}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$

Note that you **always accept** either the new or the old



Metropolis algorithm

If $q_{\text{new}|\text{old}}(x | z) = q_{\text{new}|\text{old}}(z | x)$, then this can be simplified

Metropolis sampling proceeds as follows

Given $X[0]$

For $i = 0$ to $n-1$

Generate $Z \sim q_{\text{new}|\text{old}}(\cdot | X_i)$, $U \sim \mathcal{U}[0,1]$

If $U \leq \min\left(1, \frac{q(Z)}{q(X_i)}\right)$

$X[i+1] \leftarrow Z$

Else

$X[i+1] \leftarrow X[i]$



Advantages of and Challenges in MCMC

- Samples from complicated, unnormalized densities
- Easy to implement
- Proposal density and starting point need tuning to ensure that
 - Explore all the sample space and not miss out
 - Exploit sampling in regions where the density is large



How do we measure quality of MCMC

Short answer: We cannot easily

Longer answer: Given a symmetric, positive definite kernel, $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, the **discrepancy**, sometimes called the maximum mean discrepancy between two empirical distributions is defined by

$$D^2(\{\mathbf{x}_i\}_{i=0}^{m-1}, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(\mathbf{x}_i, \mathbf{z}_j) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

Requires $\mathcal{O}(\max(m, n)^2)$ operations



What is discrepancy?

The **discrepancy** measures

- The **difference** between two probability distributions
- The **worst-case error** in approximating an expectation/integral $\mu = \mathbb{E}[f(X)]$ by a sample mean when f is in the unit ball of a Hilbert space with **reproducing kernel** K
- The **root-mean square error** in approximating an expectation/integral $\mu = \mathbb{E}[f(X)]$ by a sample mean when f is an instance of a Gaussian process with **covariance kernel** K

Discrepancies from kernels were popularized by [Hickernell 1998] and [Gretton et al. 2012]



Symmetric, Positive Definite Kernels

- A square matrix, $\mathbf{K}$, is symmetric and positive definite iff

$$\mathbf{K}^T = \mathbf{K} \quad \text{and} \quad \mathbf{c}^T \mathbf{K} \mathbf{c} > 0 \text{ for all } \mathbf{c} \neq \mathbf{0}$$

- A kernel, $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, is **symmetric and positive definite** iff

$$\mathbf{K} := \left(K(\mathbf{x}_i, \mathbf{x}_j) \right)_{i,j=0}^{n-1} \quad \text{is symmetric and positive definite}$$

for all $n \in \mathbb{N}$ and $\{\mathbf{x}_i\}_{i=0}^{n-1}$ with distinct elements, e.g.

$$K(\mathbf{t}, \mathbf{x}) = \exp\left(-\|\mathbf{t} - \mathbf{x}\|^2/h^2\right), \quad \mathcal{X} = \mathbb{R}^d, \quad \text{squared exponential}$$

$$K(\mathbf{t}, \mathbf{x}) = \prod_{\ell=1}^d \left[1 + \frac{\gamma_{\ell}^2}{2} \left(|t_{\ell} - 1/2| + |x_{\ell} - 1/2| - |t_{\ell} - x_{\ell}| \right) \right], \quad \mathcal{X} = [0,1]^d$$

centered discrepancy



Discrepancy from kernels

Suppose that

- $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a symmetric, positive definite kernel,
- $\mathcal{X}$ is the sample space for random variables/vectors $\mathbf{X}$ and $\mathbf{Z}$
- $\mathbf{X}$ has CDF F_X and $\mathbf{Z}$ has CDF F_Z

Then **discrepancy** measures the difference between two distributions:

$$\begin{aligned} D^2(F_X, F_Z, K) &= \mathbb{E}_{\mathbf{X}, \mathbf{X}' \sim F_X} K(\mathbf{X}, \mathbf{X}') - 2\mathbb{E}_{\mathbf{X} \sim F_X, \mathbf{Z} \sim F_Z} K(\mathbf{X}, \mathbf{Z}) + \mathbb{E}_{\mathbf{Z}, \mathbf{Z}' \sim F_Z} K(\mathbf{Z}, \mathbf{Z}') \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}') - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) dF_X(\mathbf{x}) dF_Z(\mathbf{z}) \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') dF_Z(\mathbf{z}) dF_Z(\mathbf{z}') \geq 0 \end{aligned}$$



Discrepancy from kernels

Then **discrepancy** measures the difference between two distributions:

$$\begin{aligned} D^2(F_X, F_Z, K) &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') dF_X(\mathbf{x}) dF_X(\mathbf{x}') - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) dF_X(\mathbf{x}) dF_Z(\mathbf{z}) \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') dF_Z(\mathbf{z}) dF_Z(\mathbf{z}') \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') d[F_X(\mathbf{x}) - F_Z(\mathbf{x})] d[F_X(\mathbf{x}') - F_Z(\mathbf{x}')] \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') \varrho_X(\mathbf{x}) \varrho_X(\mathbf{x}') d\mathbf{x} d\mathbf{x}' - 2 \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{z}) \varrho_X(\mathbf{x}) \varrho_Z(\mathbf{z}) d\mathbf{x} d\mathbf{z} \\ &\quad + \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{z}, \mathbf{z}') \varrho_Z(\mathbf{z}) \varrho_Z(\mathbf{z}') d\mathbf{z} d\mathbf{z}' \geq 0 \end{aligned}$$



Discrepancy with empirical distributions

The formula for the squared discrepancy works if one or both of the distributions is the **empirical distribution** of a sample.

$$\begin{aligned} D^2(F_X, F_{\{z_i\}_{i=0}^{n-1}}; K) &= D^2(F_X, \{z_i\}_{i=0}^{n-1}; K) \\ &= \int_{\mathcal{X} \times \mathcal{X}} K(x, x') dF_X(x) dF_X(x') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(x, z_i) dF_X(x) \\ &\quad + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(z_i, z_j) \end{aligned}$$

$$D^2(\{x_i\}_{i=0}^{m-1}, \{z_i\}_{i=0}^{n-1}; K) = \frac{1}{m^2} \sum_{i,j=0}^{m-1} K(x_i, x_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(x_i, z_j) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(z_i, z_j)$$



Unbiased estimates of squared discrepancy

The squared discrepancy is non-negative, but we can construct **unbiased and possibly negative** estimates of $D^2(F_X, F_Z, K)$ based on IID samples $\{\mathbf{x}_i\}_{i=0}^{m-1}$ and/or $\{\mathbf{z}_i\}_{i=0}^{n-1}$

$$D_{\text{unb}}^2(F_X, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}') d\varrho_X(\mathbf{x}) d\varrho_X(\mathbf{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{z}_i) d\varrho_X(\mathbf{x}) \\ + \frac{1}{n(n-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$

$$D_{\text{unb}}^2(\{\mathbf{x}_i\}_{i=0}^{m-1}, \{\mathbf{z}_i\}_{i=0}^{n-1}; K) = \frac{1}{m(m-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{m-1} K(\mathbf{x}_i, \mathbf{x}_j) - \frac{2}{mn} \sum_{i,j=0}^{m-1, n-1} K(\mathbf{x}_i, \mathbf{z}_j) \\ + \frac{1}{n(n-1)} \sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} K(\mathbf{z}_i, \mathbf{z}_j)$$